



Anubhav Jha <anubhavjha0812@gmail.com>

Assignment for role of Quant Developer

2 messages

Purnima Sinha <head.hr@veradevelopers.com>
To: Anubhav Jha <anubhavjha0812@gmail.com>

26 September 2026 at 18:24

Dear Anubhav,

Thank you for showing interest with us. As requested, please find below the details of the technical test as shared:

Test Instructions:

Develop primarily with Claude Code and use the below-mentioned instructions:

- Extend and maintain the live grid and stop-and-reverse execution engines — ATR-based spacing, pyramiding, kill switches, position caps.
- Build out the technical analysis module: trend, momentum, volatility and volume indicators, with a single tested implementation each and no duplicated math.
- Wire the Macro Regime Engine: ingest macro proxies, score them into regime states, apply parameter overrides, enforce circuit breakers.
- Own broker and data integration — Zerodha Kite Connect (REST + WebSocket), tick vendors, MCX and NSE contract masters, expiry and rollover handling.
- Build and maintain the backtest harness: bar-accurate fills, slippage and cost modelling, walk-forward, no lookahead. Backtest numbers must reconcile to live numbers, and both must reconcile to the desk's spreadsheets.
- Order and state management: idempotent placement, order-state reconciliation after restarts, position and P&L truth, crash recovery.
- Observability: structured logs, trade blotters, and alerting to phone the moment the engine deviates from expectation.
- Write tests. Every strategy change ships with a regression test that would have caught the bug.
- Make agents to automate the SDLC.

Must Have:

- **Python (2+ yrs, production):** pandas, numpy, dataclasses, type hints, virtualenv/poetry, pytest. You know why `iterrows()` in a hot loop is a problem and when vectorisation is actually the wrong answer.
- **Concurrency:** asyncio or threading in anger — WebSocket consumers, queues, back-pressure, graceful shutdown. You have debugged a race condition you caused.
- **API integration:** REST + WebSocket, auth/token refresh, rate limits, retries with backoff, idempotency, reconnect-and-resync.
- **Indian market plumbing:** MCX contract specs (lot size, tick size, price quotation basis, expiry, staggered delivery), NSE F&O basics, margins, STT/CTT and brokerage mechanics.
- **Numerical discipline:** you treat "matches to the paisa" as the acceptance criterion, not "roughly matches".
- **Git:** meaningful commits, code review as a normal activity.

Good to Have:

- Live trading experience with your own or someone else's capital.
- TA-Lib / pandas-ta, vectorbt or backtrader, and an opinion on their limits.
- SQL / time-series storage (Postgres, TimescaleDB, DuckDB, Parquet), Redis, Docker.
- Streamlit / FastAPI for internal dashboards; Grafana for monitoring.
- Exposure to tick-level data pipelines (TrueData, GDFL or similar)

Note: Please submit the assignment by October 1, 2026.

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**Best Regards,
Purnima Sinha**

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Anubhav Jha <anubhavjha0812@gmail.com>

26 September 2026 at 18:28

To: Purnima Sinha <head.hr@veradevelopers.com>

Thank you for sharing the assignment details, will complete the assignment by October 1.

Best Regards,

Anubhav Jha | [LinkedIn](#)

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